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Experimental Investigation on the Surface and Volume Homogeneity of Additive Manufactured Fused Silica Components in the Laser Glass Deposition Process

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Abstract

Laser Glass Deposition is an additive manufacturing method to produce individualized structural components out of glass. A CO_2 laser is utilized as a heat source to melt fused silica filaments and transform them into a formable viscous state. The fiber filament is fed laterally under a defined angle into the process zone. The viscous filament is deposited layer-by-layer using a 3-axis linear system with an integrated rotational axis. To investigate the surface and volume quality of the additively manufactured fused silica components, fully dense test specimens are analyzed in this paper. Quality characteristics such as surface roughness, formation of boundary layers and optical transparency constitute the focus of the investigations. Consequently, fully dense glass components with homogeneous volume structures without pores and boundary layers and a surface roughness of less than 30 nm were printed successfully.

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Keywords: Glass based additive manufacturing; Laser material processing; Fused silica; Free-form optics; Layer-free 3D printing

1. Introduction

Silicate glasses play an important role in industrial use of glasses due to their optical, chemical, and thermal properties [1]. Fused silica in particular is distinguished by its low thermal expansion coefficient and high thermal shock resistance [2]. It is used for a variety of optical components and optical fibers, but also finds application in chemical and medical apparatus engineering as well as in semiconductor technology [3]. However, the processing temperatures of over 2000 °C pose challenges in handling the low-viscosity fluid to form complex or highly individualized components.

Thanks to its freedom of design and process, the additive manufacturing of glass enables the realization of special and customized solutions with comparatively little effort compared to conventional manufacturing processes. Several approaches for the additive manufacturing of glass materials can be found in the literature. Xin et al. gave a comprehensive overview of the current state of research into additive glass processing methods in [4]. These can be broadly categorized into the following:

- Sinter-based processes [5]
- Stereolithography processes [6]
- Extrusion processes [7]
- Deposition welding processes [8][9]

In the investigations presented here, laser glass deposition (LGD), which is one of the deposition welding processes, is used to produce test specimen to analyze the surface quality

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and volume homogeneity. For many applications of glass components, a smooth surface and a transparent volume are important quality features.

2. Methods

The experimental setup (Fig. 1) consists of a CO_2 laser (Diamond J3, Coherent Inc.), a 4-axis system (Jenaer Antriebstechnik GmbH), a fiber feeding unit (MiniStirrer, HiTec Zang GmbH) to feed fused silica filament material (Weinert Optics GmbH) and the quartz glass substrate which serves as printing bed (Schröder Spezialglas GmbH).

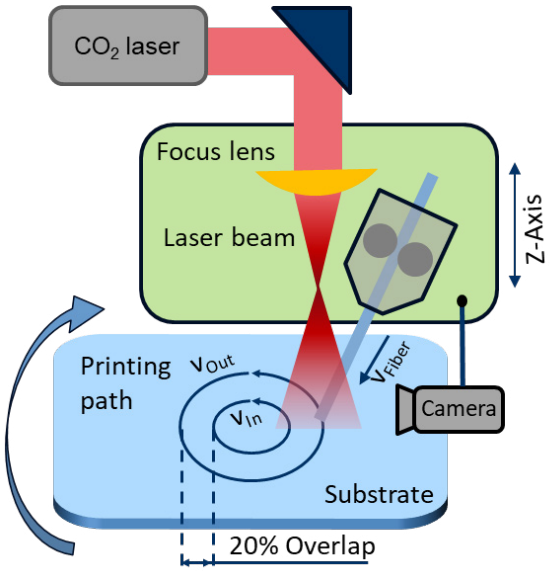


Figure 1: Schematic sketch of the experimental set up.

The radiation of the CO_2 laser (P_L = 200 W) is used as a heat source for melting and fusing the fused silica glass fiber filament (800 µm diameter) onto the quartz glass substrate. The laser beam diameter (d_{Laser}) is defocused to form a beam diameter of 7 mm. The fiber is introduced laterally into the generated process zone at a rate of v_{Fiber} = 90 – 396 mm/min, and an angle of 55°. The glass melt can be deposited on the glass surface in any shape and contour using the 3 linear and a rotational axis. A camera (IDS Imaging Edmund Optics Ltd) is used for monitoring the deposition of the glass filament. The fiber feeding unit, the focus lens and the camera are attached to the Z-axis so that a new layer can be deposited under the same process conditions.

In the following investigations, cylindrical test specimens are printed to analyze the surface and volume quality. In order to print a dense glass volume, two concentric rings are printed with an overlap of 20% to each other. The inner ring is printed at a speed of v_{Inside} = 1 rpm and the outer ring at $v_{Outside}$ = 4 rpm. The printing path can be seen in principle in Fig 1 and is illustrated again in detail in Fig 2 (a).

3. Results

3.1 Contour Study

To analyze the surface quality and investigate the influence of the test parameters, cylindrical test specimens were manufactured with different fiber feed rates v_{Fiber} = (90 – 396) mm/min. Exemplary images of the test specimens for the respective feed rates can be seen in Fig. 2 (b). An optically visible increase in surface quality can be observed in Fig. 2 (b) with a decreasing feed rate. For the highest value of v_{Fiber} = 396 mm/min, a wavy surface contour can be observed along the complete structure. At a reduced feed rate, the individual layers can be fused together better, so that the wavy part of the surface shrinks. For v_{Fiber} = 180 mm/min the waviness was less prominent, while for v_{Fiber} = 126 mm/min a complete smooth area of 8 mm high was achieved. For v_{Fiber} = 90 mm/min the 10 mm high specimen was printed smoothly without any waviness. With the reduced feed rate, however, an increase in the flank angle can also be observed, which is graphical represented in Fig. 3. With the highest feed rate, a good approximation to the target contour can be observed, with a flank angle of 2.15°. With the lowest value of v_{Fiber} = 90 mm/min a flank angle of 14.04° +/- 0.93° can be observed. The slower feed rate reduces the mass flow rate supplied, which leads to increased heat input at constant P_L, causing the molten glass to spread further. As the laser spot diameter is larger than the overlap zone, the layers slowly fall into each other as the number of layers increases. At high feed rates, this effect is not significant, as the high volume flow in the process zone reduces the heat input, resulting in a higher viscosity and a lower surface tension that influences the melting dynamics.

3.2 Surface Roughness Study

To analyze the surface quality of the additively manufactured test specimens, the surface roughness was determined exclusively on the smooth areas without boundary layers. It is important to note that the samples were not prepared by subsequent polishing but were measured directly from the printing process. The analysis was carried out using a tactile profile measuring device MarSurf LD 130 from Mahr GmbH. The measurements were carried out in accordance with the standards DIN EN ISO 1661021 and DIN EN ISO 4287 for surface roughness measurements to ensure comparability with industrially manufactured glass components. The roughness profile of an example measurement for v_{Fiber} = 90 mm/min can be seen in Fig. 2 (c). An arithmetic roughness of 21 nm was determined for a measuring distance of 1.26 mm. The corresponding arithmetic roughness of the remaining test specimens are shown graphically in Fig. 4. A single specimen is measured at six positions to calculate an average value. This is carried out on four further test specimens of the same process parameters to be able to represent a standard deviation in the form of the error bars in Fig 3.

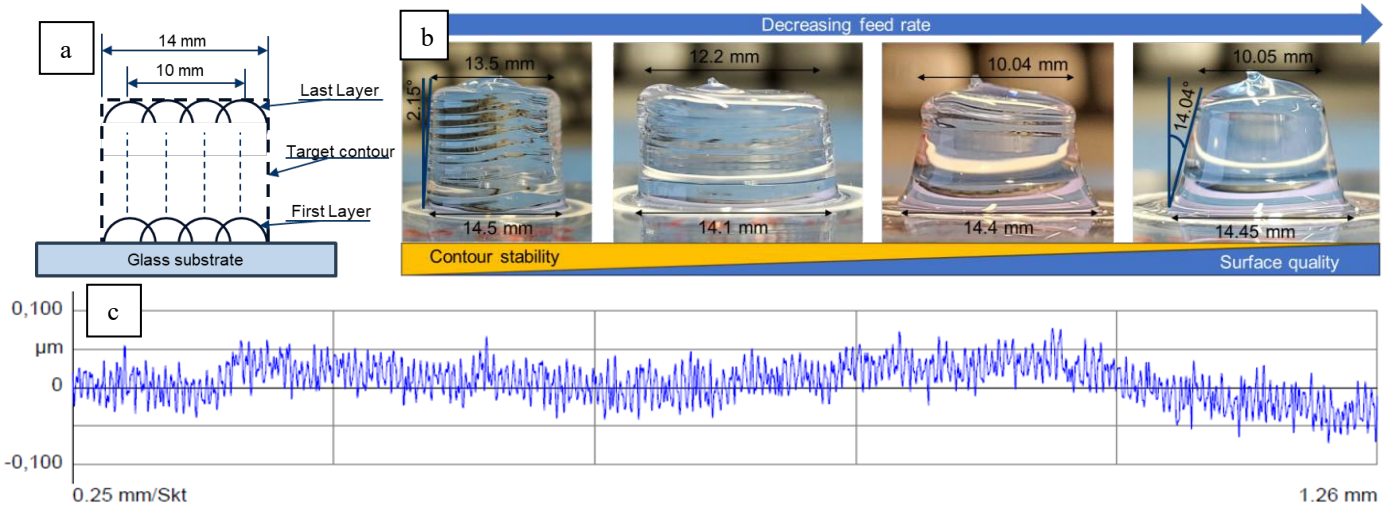


Figure 2: a: Schematic sketch of the target contour for the test specimen. The diameter of the outer circle is set up at 10 mm from center to center of the single track. The height is not set up on a specific value. b: Images of the additively manufactured test specimen. The feed rate was decreased from left to right (396 to 90 mm/min). While the surface and optical quality is increasing, the contour quality is decreasing, symbolizing through the increased flank angle. c: Surface roughness measurement for v_{Fiber} = 90 mm/min performed with perthometer according to DIN EN ISO 1661021 and DIN EN ISO 4287. The measured arithmetic roughness is 21 nm.

Roughness's of 20 - 40 nm can be observed for v_{Fiber} = 90 – 126 mm/min. From v_{Fiber} = 180 mm/min, an increased arithmetic roughness of 220 nm is measured, which is caused by the greater unevenness (as can also be seen in Fig. 2 (b). For v_{Fiber} > 180 mm/min the waviness becomes too significant to measure a reliable roughness value. The small roughness's in the remaining measurements are caused by the complete fusion of the individual layers, whereby the surface is formed due to the cohesive forces of the low-viscosity glass melt. Based on the small standard deviations of the measurements (+/- 5.6 nm), it can be concluded that reproducible sample bodies can be produced by the process.

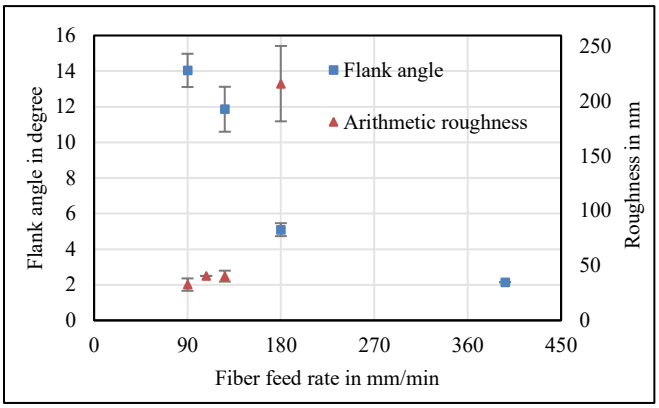


Figure 3: Graphical representation of the flank angle and the surface roughness as a function of the feed rate. The error bars represent the standard deviation of the measurements.

3.3 Optical and Volume Quality Study

To analyze the optical quality and volume homogeneity of the additive manufactured specimens, a beam profile measurement was carried out. The setup of the beam profile measurement is shown in Fig. 4. A He-Ne laser (632 nm) is used as the beam source, whose radiation is magnified and collimated to a beam diameter of 2 mm by two focusing lenses. This beam is incident vertically on the underside of the clamped specimen so that the radiation is first propagated through the substrate and then through the specimen and finally detected in the beam camera. The beam camera consists of a CCD camera, which is controlled by a Raspberry Pi 4b and outputs the detected light as a false color live image. The upper side of the test specimen was remelted after the printing of the last layer by rotating the specimen under CO_2 laser radiation to obtain a homogeneous surface. The entire specimen can be seen in Fig. 5 (b) and a view from above through the complete specimen can be seen in Fig. 5 (c) already suggesting a bubble-free volume.

Fig. 5 (a) shows the beam profile measurement at a distance of 30 mm from the focus. It can be observed that the beam profile has an elliptical shape and thus deviates from the circular input profile. The explanation is that due to the additive manufacturing process, no rotationally symmetrical sphere has formed on the surface, but the curvature is more pronounced on one side than on the other. What is remarkable, however, is the homogeneity of the beam profile, which shows neither bubbles nor boundary layer-related distortions for an additively manufactured component. This is a clear indication that the printed material has no boundary layers either in the transition area from the substrate to the printed material or in the internal layer transitions. The results show that targeted beam shaping is also possible with a suitable printing strategy for the surface contour.

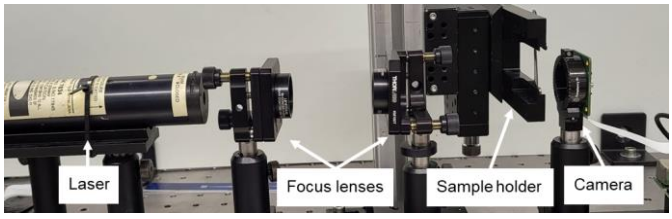


Figure 4: Photography of the measurement set up for beam profile detection.

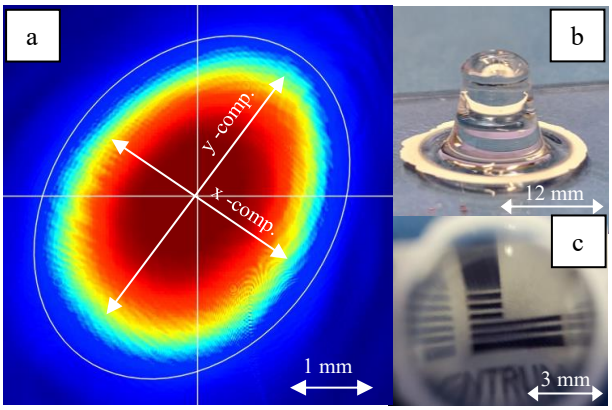


Figure 5: a: Beam profile measurement through an additive manufactured test glass specimen. The upper surface was remelted at the end of the printing process. b: Image of the manufactured specimen. c: Top view through the printed structure.

The respective beam components in the x/y direction for different distances from the focus are shown graphically in Fig. 6. The respective beam components show a linear increase with larger distance, which underlines the optical function of the printed structure. It can also be observed that the distance between the x and y components increases equally, emphasising the spherical nature of the surface curvature.

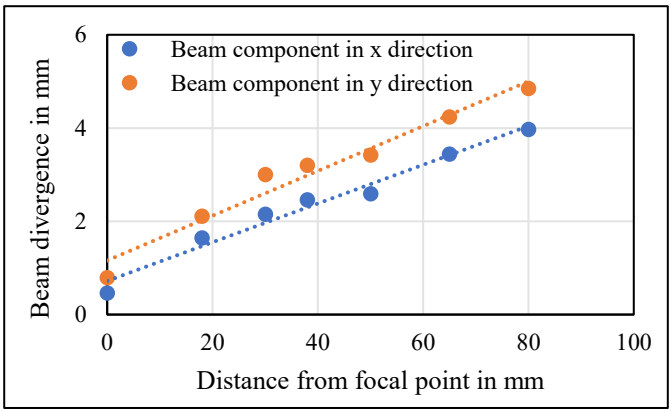


Figure 6: Graphical representation of the x/y components of the beam diameters for different distances from the focal point.

4. Conclusion

Additive manufactured test specimens were produced using Laser Glass Deposition and analysed to investigate the surface and volume quality. The cylinder-shaped test specimens were printed by depositing two concentric rings with an overlap of 20%. The printing parameters (P_L = 200 W, v_{Inside} = 4 rpm, $v_{Outside}$ = 1 rpm, d_L = 7 mm) were kept constant except for v_{Fiber} = (90 , 108, 126, 180, 396) mm/min. The resulting component contour deviated more strongly from the target contour as the feed rate decreased. The slower feed rate increased the heat input into the process zone, which led to increased flank formation on the outer contour. On the other hand, the increased heat input allowed the individual layers to fuse together completely, resulting in a smooth surface free of waves and boundary layers. The surface roughness of the smooth areas of each specimen was measured. The arithmetic roughness of the specimens was measured at different points for several specimens along a measuring distance of 1.26 mm. The lowest arithmetic roughness was determined for v_{Fiber} = 90 mm/min at 21 nm. Furthermore, the volume homogeneity and optical quality were demonstrated by a beam profile measurement. The transmitted beam showed a slight ellipticity, but was generally characterised by a homogeneous intensity distribution, which was free of bubble or volume boundary layer distortions.

The results show the strengths and challenges of the LGD method. It is possible to produce dense, smooth and boundary layer-free glass components with arithmetic roughness's in the nanometer range. This opens up the possibility of extending the limits of modern glass production and realising components with new or integrated functions. The main challenge lies in contour accuracy and contour retention to form individulized optical functionalities, which will be focused on in future studys.

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